

Day 4 User Authentication & Spring Security

Date

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Overview

Implemented user management and Spring Security authentication. Added role-based user entity with BCrypt password encryption, registration/login REST APIs, and Thymeleaf login/register pages.

Tech Stack

- Java 26
- Spring Boot 4.1.0
- Lombok 1.18.46
- MongoDB (clinic_db)
- Thymeleaf + Bootstrap 5
- Spring Security
- BCrypt Password Encoder

Files Created

New Java Files

1. `com.clinic.enums.Role` - Enum with ADMIN, DOCTOR, RECEPTIONIST
2. `com.clinic.entity.User` - User document with `userId`, `username`, `email`, `password`, `role`, `createdAt`
3. `com.clinic.dto.RegisterRequestDto` - DTO for user registration
4. `com.clinic.dto.LoginRequestDto` - DTO for user login
5. `com.clinic.dto.UserResponseDto` - DTO for user response (excludes password)
6. `com.clinic.exception.UserNotFoundException` - Custom exception for user errors
7. `com.clinic.repository.UserRepository` - MongoDB repository for users
8. `com.clinic.service.UserService` - Interface for user operations
9. `com.clinic.service.UserServiceImpl` - Implementation with BCrypt encoding
10. `com.clinic.security.CustomUserDetailsService` - Spring Security `UserDetailsService`
11. `com.clinic.config.SecurityConfig` - Spring Security filter chain config
12. `com.clinic.config.MongoConfig` - Explicit MongoDB connection to `clinic_db`
13. `com.clinic.controller.AuthRestController` - REST endpoints for auth
14. `com.clinic.controller.AuthViewController` - Thymeleaf auth pages

New HTML Files

1. `templates/auth/login.html` - Bootstrap login form
2. `templates/auth/register.html` - Bootstrap registration form with role dropdown

Updated Files

1. `pom.xml` - Added `spring-boot-starter-security` dependency
2. `templates/index.html` - Added Users card, Login/Register nav links
3. `templates/patients/list.html` - Added Login/Register to navbar

4. templates/patients/form.html - Added Login/Register to navbar
5. templates/patients/detail.html - Added Login/Register to navbar
6. templates/doctors/list.html - Added Login/Register to navbar
7. templates/doctors/form.html - Added Login/Register to navbar
8. templates/doctors/detail.html - Added Login/Register to navbar

REST API Endpoints

Method	Endpoint	Description
POST	/api/auth/register	Register new user with BCrypt password
POST	/api/auth/login	Login user (returns user data)
GET	/api/auth/me	Get user by username

Thymeleaf Pages

URL	Description
/login	Login form
/register	Registration form with role selection

Key Features Implemented

1. User entity with custom USR-XXXX ID format
2. Role enum: ADMIN, DOCTOR, RECEPTIONIST
3. BCrypt password encryption (passwords not stored in plain text)
4. Duplicate username/email validation
5. Spring Security form login integration
6. CustomUserDetailsService for loading users from MongoDB
7. Public access to home, patients, doctors, login, register pages
8. Error handling on registration form

Issues Faced & Solutions

Issue 1: Spring Security blocking form submission

- Error: 403 Forbidden on /auth/register
- Solution: Added /auth/** to permitAll() in SecurityConfig

Issue 2: Login redirecting to /login?error

- Error: UserDetailsService not found
- Solution: Created CustomUserDetailsService implementing UserDetailsService

Issue 3: Users collection created in test database instead of clinic_db

- Error: MongoDB defaulting to test database
- Solution: Created explicit MongoConfig.java with MongoTemplate("clinic_db")

Testing

REST API Test (Postman)

POST /api/auth/register { "username": "admin", "email": "admin@clinic.com", "password": "admin123", "role": "ADMIN" } Response: 201 Created with user data

Web Form Test

1. Visit /register
2. Fill form with username, email, password, role
3. Submit -> Redirects to /login
4. Login with credentials -> Redirects to homepage